



DOCUMENT AWARE CONVERSATIONAL RAG

● TEAM NAME : MEDHAVI

EVERY QUERY, ONE ANSWER

WHY ?

- 01** McKinsey found interaction workers spend 28% of the week managing email and nearly 20% looking for internal information
- 02** NBER found access to a generative AI assistant increased worker productivity by about 14%, with larger gains for less-experienced workers
- 03** Gartner says poor data quality costs organizations at least USD 12.9 million per year on average
- 04** Microsoft found 62% employees struggle with too much time spent searching for information.

RAG HELPS WITH

FINANCIAL LOSSES REDUCTION
AND OPERATIONAL EFFICIENCY
IMPROVEMENT

Simplifies email or document management.

AI-based assistance can produce measurable productivity improvement.

Grounded Answers : Cited answers from the source.

INNOVATION

DUAL MODE DOCUMENT UNDERSTANDING

Processes both text and visuals — scanned PDFs, charts, engineering diagrams, and images are all searchable, not just typed documents. Built using ColPali + Mistral OCR + Llama Vision.

DUAL INDEXING FOR MAXIMUM RECALL

Uses Raptor Hierarchical Indexing alongside Chunk + Summary indexing — meaning it understands both the big picture of a document and its granular details simultaneously.

HYBRID RETRIEVAL + RERANKING

Uses Raptor Hierarchical Indexing alongside Chunk + Summary indexing — meaning it understands both the big picture of a document and its granular details simultaneously.

CRAG VERIFICATION LOOP

It Knows What It Doesn't Know Every answer goes through a self-verification loop using Score Chunks, Support Check, and Lettuce Detect. If the answer isn't found, it says "Not Found" instead of hallucinating — critical for compliance and legal use cases at Adani.

COMPARSION

VS STATE OF THE ART / LLMS :

OUTPUT STYLE

Can output whole document abstractive summary but not very niche cited details, specially from Infographics.

SOURCE GROUNDING

LLM output based of off learned weights and parameters and Tend to Hallucinate outside of the specified sources.

DOCUMENT DEPTH AND LIMITATIONS

Can generate polished summaries and explanations, but answers are limited by input context and may not be reliatbly cite exact internal source.

VS GENERIC RAG :

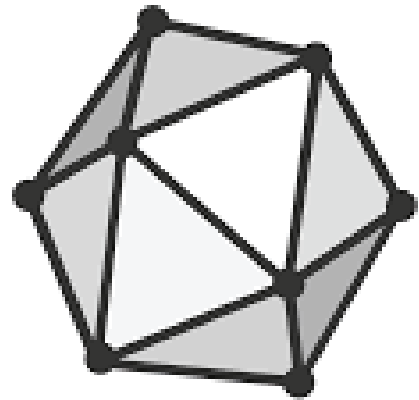
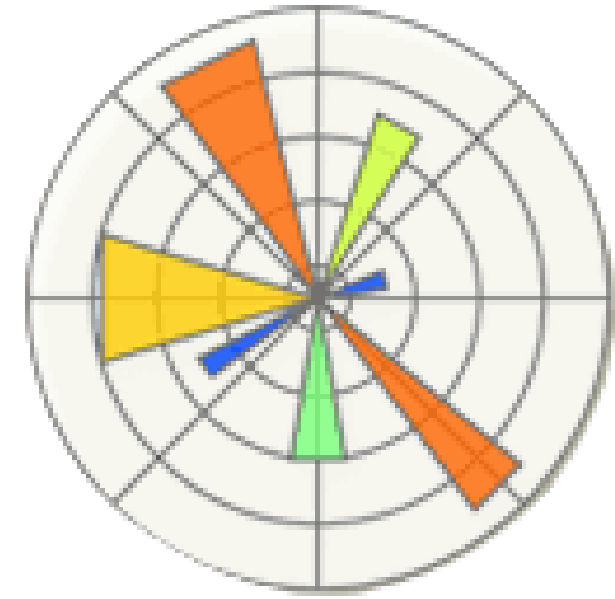
OUTPUT STYLE

Can output specific details not neccesarily at infographics level but is unable to generate document level summaries.

OCR HEAVY

Generic pipelines are text/OCR heavy, so tables, charts , diagrams , page layout and infographics can lose meaning.

TECH STACK

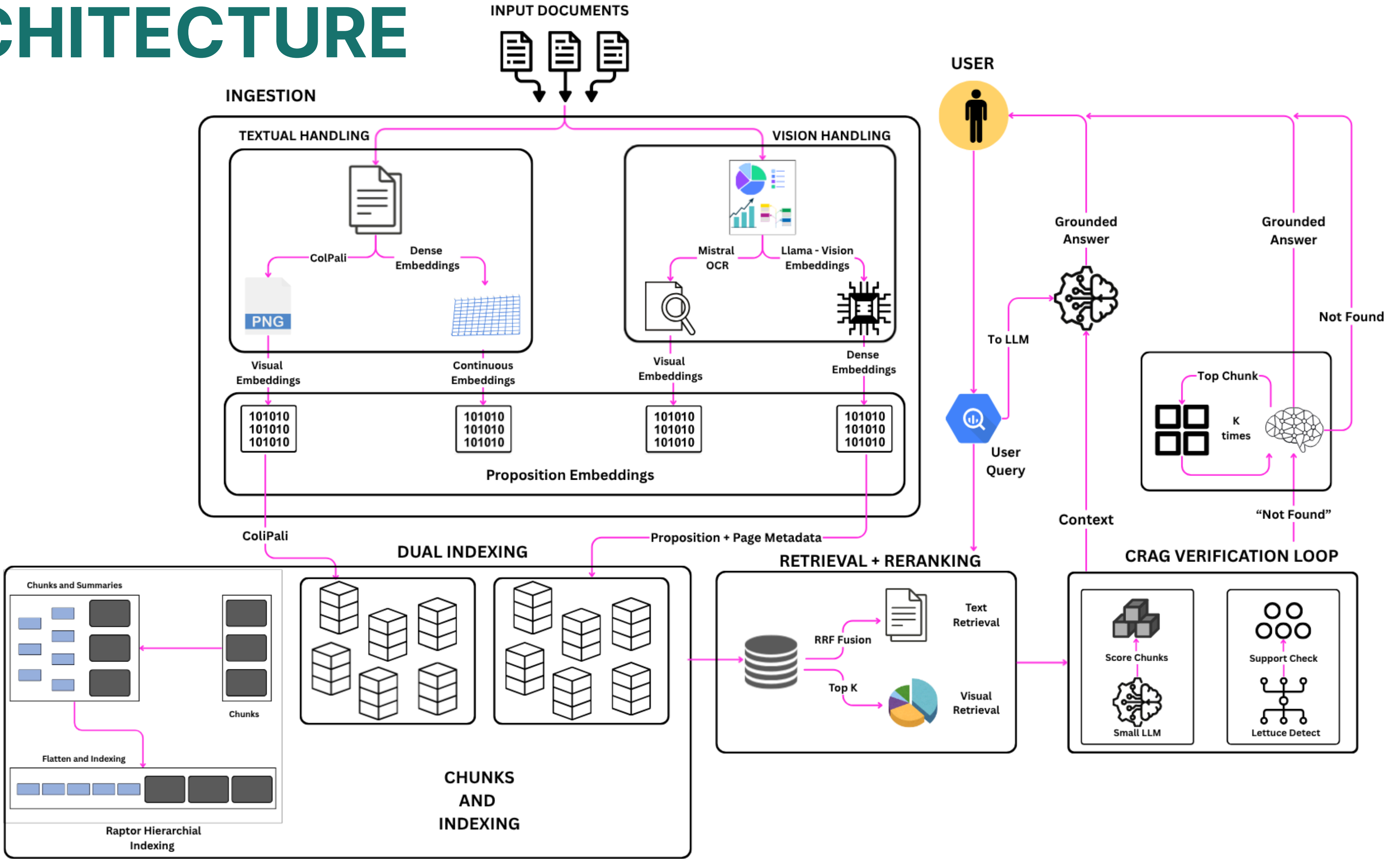


Hugging Face



Weights & Biases
by CoreWeave

ARCHITECTURE



PARAMETERS

CHUNK SIZE	512	CHAR	2400 in One Chunk
TEXT CHUNKS	408	VLM	185
VLM CHUNKS	185	SIZE	8 in 1 Block
BLOCK SIZE	73	LVL	4

SPECIFICATIONS

QWEN-2.5-VL	for visual embeddings
COLPALI	Image based understanding
LLAMA3.3-8B	Output Generation
MINIVLM	for Text Embeddings
MISTRAL OCR	OCR Image & Table Recognition
E5 SMALL	Propositional Embeddings
GEMINI2.5FLASH	FALLBACK FOR QWEN-2.5-VL



BUSINESS/ VALUE CREATION

Search-time reduction

Employees waste time finding the right PDF, policy, SOP, financial report, or clause.

RAG retrieves relevant chunks/pages and gives direct cited answers.

Faster decision-making

Teams wait for SMEs or managers to locate documents. Users can ask questions directly and get grounded answers from internal files.

Lower Rework

Wrong version of a document or missed clause can cause repeated work. Source-grounded retrieval reduces reliance on memory or outdated files.

Operational Troubleshooting

Maintenance, plant, port, logistics, and field teams need fast access to SOPs/manuals. Multimodal retrieval handles text, tables, charts, diagrams, and page images.

Scalable deployment

Containerized services can run across environments; Kubernetes is designed for deployment, scaling, and management of containerized applications.

Knowledge retention

system turns historical documents into institutional memory.

FINANCIAL IMPACT

Poor data quality / wrong information

Employees waste time finding the right PDF, policy, SOP, financial report, or clause.

RAG retrieves relevant chunks/pages and gives direct cited answers.

Poor project execution

PMI reports 11.4% of investment is wasted due to poor project performance.

Industrial downtime

Maintenance, plant, port, logistics, and field teams need fast access to SOPs/manuals. Multimodal retrieval handles text, tables, charts, diagrams, and page images.

Employee search waste

McKinsey says nearly 20% of workweek goes into looking for internal information or people who can help.

REAL WORLD RAG EXAMPLES

Morgan Stanley

Morgan Stanley built a GPT-4-based assistant for their over 16,000 advisors. With it, advisors instantly search over 100,000 internal documents to automatically generate client-meeting debriefs, reports, and action items.

Ballard Spahr

Saved \$2M in unplanned losses, cut proposal prep by 2 hours per proposal, reduced nonbillable research time by 60%

Klarna

In first month: 2.3 million conversations, handled two-thirds of customer-service chats, equivalent to 700 full-time agents, repeat inquiries down 25%, resolution time reduced from 11 minutes to under 2 minutes, available in 23 markets / 35+ languages, estimated \$40M profit improvement in 2024

ADANI USECASES

Unified search across all internal reports, SOPs, policies, safety manuals, compliance docs, project files, and financial documents — spanning Adani's entire ₹40,000 Cr project portfolio where every project operates with its own siloed documentation.

6 out of 10 listed Adani companies received SEBI show-cause notices for non-compliance with listing regulations in FY2024 — directly showing the cost of fragmented legal document management

Automate repetitive queries from 350M users across airports, 120M+ MMT cargo from ports, energy, gas, HR, IT, and procurement queries using only verified company documents.

COMPARISON

Adani Enterprises Q2 FY26 Earnings Presentation Highlights

Create notebook

Analytics

Share

Settings

PRO

Sources

AEL_Earnings_Presentation_Q2-FY26_copy.pdf

Source guide

mining sectors toward rapidly growing ventures in airports, green hydrogen, and data centres. Key financial metrics reveal a resilient EBITDA, driven by the expansion of incubating businesses, while identifying massive upcoming projects like the Navi Mumbai Airport and Ganga Expressway as future value drivers. Ultimately, the presentation aims to reassure investors of the group's financial stability and its proven ability to transform emerging industries into profitable, world-class infrastructure assets.

Financial performa...

Business incubat...

Infrastructure developm...

Adani portfolio overvi...

Green hydrogen ecosyst...

Today • 23:39

In the Adani Portfolio overview infographic on page 6, under which exact parent category column is 'NDTV' placed, and what is the exact promoter holding percentage shown next to it?

In the Adani Portfolio overview infographic, **NDTV** is placed under the **Direct Consumer** parent category column 1 2 .

The exact promoter holding percentage shown right next to it is **64.71%** 1 .

(Note: A related footnote for this infographic mentions that the actual promoter holding in NDTV increased to 69.02% following the completion of a rights issue in October 2025 3 .)

Would you like to know more about the other businesses in the Adani portfolio or their financial highlights?

Save to note

Start typing... 1 source

Studio

Create an Audio Overview in: हिन्दी, বাংলা, ગુજરાતી, ಕನ್ನಡ, മലയാളം, मराठी, ਪੰਜਾਬੀ, தமிழ், తెలుగు

Audio...

Slide deck

Video...

Mind Map

NotebookLM can be inaccurate; please double-check its responses.

'POST' \n ://127.0.0.1:8000/chat/v2/sessions/13125699-dc6f-42c1-8c79-73501d3d4a5e/messages' \n cept: application/json' \n ontent-Type: application/json' \n ent": "In the Adani Portfolio overview infographic on page 6, under which exact parent category column is '\n\nNDTV\n\n' placed, and what is the exact promoter holding percentage shown r

RL

127.0.0.1:8000/chat/v2/sessions/13125699-dc6f-42c1-8c79-73501d3d4a5e/messages

ponse

Details

Response body

```
e shown next to it?",
  "meta": null,
  "created_at": "2026-05-03T18:00:44.161760"
},
"assistant_message": {
  "id": "4453501b-e8e1-4ad0-843a-6885334ae433",
  "session_id": "13125699-dc6f-42c1-8c79-73501d3d4a5e",
  "role": "assistant",
  "content": "According to the context [p6:c38], 'NDTV' is placed under the 'Other' category, and the exact promoter holding percentage shown next to it is 64.71% Adani Family equity stake.",
  "meta": {
    "is_fallback": false,
    "sources": [
      {
        "chunk_id": "doc_52829a34df8b6a71:p6:c39",
        "document_id": "doc_52829a34df8b6a71",
        "filename": "ba3e2b0bd22a473c9bbe249938106c02_AEL_Earnings_Presentation_Q2-FY26_copy.pdf",
        "page_number": 6,
        "chunk_index": 39,
        "score": 0.598479271,
        "citation": "[p6:c39]",
        "text": "indicating Listed cos) - 20.00% Adani Family equity stake. * Additional entities within this category (no direct connections shown from NDTV or AWL): * GCC (Li
ght blue box) - 100% AEL equity stake (implied by position under AEL's vertical line). * Specialist Manufacturing* (light blue box) - 100% AEL equity stake (implied by position u
nder AEL's vertical line). **Legend:** * A light blue box with a dark blue border indicates \"Listed cos\"."
      }
    ]
  }
},
```

Download

WORKING DEMO

Sources

3 documents available

Add sources

Search saved sources

Select all

0 selected

AEL Press Release Q2 FY26_co...

0 Bytes · 04 May 2026

Ready

Preview

Remove

AEL_Earnings_Presentation_Q...

0 Bytes · 04 May 2026

Ready

Recent sessions

5 loaded from history

The ANIL slide shows a manufacturing facility image ...

એરટીએલ સડક નિર્માણ પ્રવૃત્તિ ટેબલ મુજબ, બીજી તિમાહી 2026 માં લેન-કિમી ...

16 messages

The ANIL slide shows a manufacturing facility image ...

Solar panels appear to be manufactured there based on the ...

2 messages

PowerMind — Document Q&A

No documents selected — select sources to filter, or search across all

The ANIL slide shows a manufacturing facility image — what type of product appears to be manufactured there based on the visual?

04:17 AM

Solar panels appear to be manufactured there based on the visual

p14 · c145

.

> 10 sources retrieved

04:18 AM

According to the ARTL Roads construction activity table, what was the lane-km construction volume in Q2-26 and what was the year-on-year change?

04:21 AM

The lane-km construction volume in Q2-26 was 456.1 and the year-on-year change was 153%

p19 · c162

.

Ask a question about your documents...

Press

Enter

 to send ·

Shift+Enter

 for new line

FUTURE SCOPE

01 Retrieval Upgrades

Cross-encoder reranking + HyDE for smarter, context-aware chunk selection.

02 Scale to Enterprise

Async workers + streaming responses to handle multiple users with near-zero wait time.

03 Concurrency & Throughput

Cloud vector DB + queue-based ingestion

WHY TRUST US ?

KANTIKA : RAG BASED DERMATOLOGY CHATBOT

Accepted at GlobalNLP 2025

A workshop on Beyond English: NLP for All Languages in the Era of LLMs, associated with RANLP 2025.

What Kantika Does:

A dermatology-focused Question Answering framework designed to generate accurate, clinically aligned, and multilingual medical responses.

Core Techniques:

- *IR-CoT: Iterative reasoning for better answer formation*
- *RAPTOR-Augmented Retrieval: Hierarchical retrieval for stronger context*
- *RAG-Style Interleaved Generation: Grounded response generation from retrieved evidence*

Impact:

Kantika demonstrates how domain-specialized RAG systems can enhance healthcare AI, making it more trustworthy, practical, multilingual, and clinically useful.



THANK YOU

● TEAM MEDHAVI

